Profanity detected 👀

Look at this nice web page

Spot is mad

He's mad because he's a nice dog and he doesn't like profanity

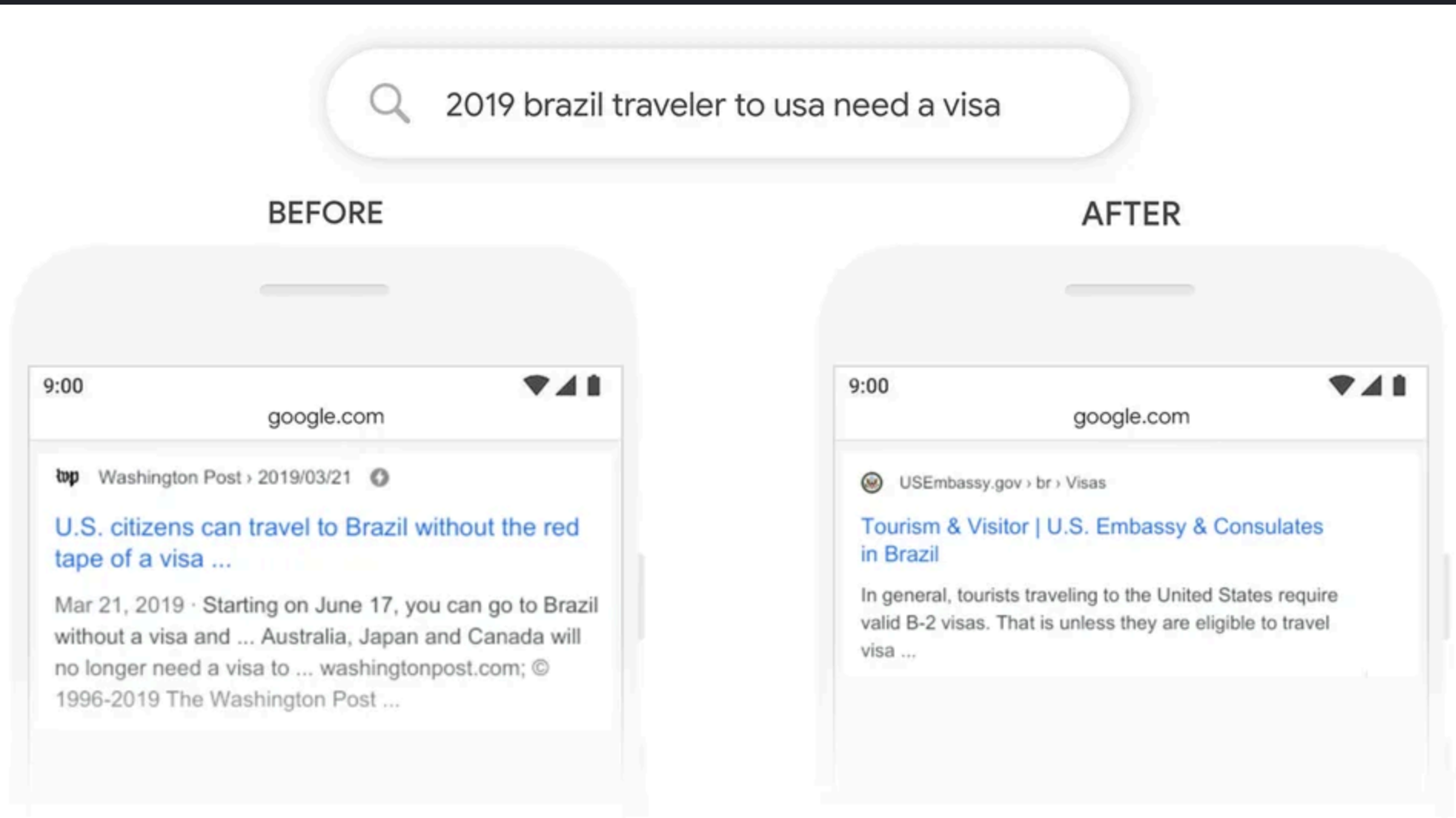
How does he know?

Google Bert


Google

Bert is a language model developed by Google:

- NLP model
- Perfect for sentiment analysis to process words and relation between them


2019 brazil traveler to usa need a visa
BEFORE
9:00
google.com
Washington Post › 2019/03/21
U.S. citizens can travel to Brazil without the red tape of a visa ...
Mar 21, 2019 · Starting on June 17, you can go to Brazil without a visa and ... Australia, Japan and Canada will no longer need a visa to ... washingtonpost.com; © 1996-2019 The Washington Post ...
AFTER
9:00
google.com
USEmbassy.gov › br › Visas
Tourism & Visitor | U.S. Embassy & Consulates in Brazil
In general, tourists traveling to the United States require valid B-2 visas. That is unless they are eligible to travel visa ...

Some code

Training

```python
def train(
    model: BERTClassifier,
    data_loader: DataLoader,
    optimizer: AdamW,
    scheduler,
    device,
    index,
):
    model.train()

    for batch in tqdm(data_loader):
        optimizer.zero_grad()

        input_ids = batch["input_ids"].to(device)
        attention_mask = batch["attention_mask"].to(device)
        labels = batch["label"].to(device)

        outputs = model(input_ids=input_ids, attention_mask=attention_mask)

        loss = torch.nn.CrossEntropyLoss()(outputs, labels)
        loss.backward()

        optimizer.step()
        scheduler.step()

    # Save model checkpoint after each batch
    torch.save(model.state_dict(), f"/data/model_checkpoint_{index}.pth")
```

```
model.train()
```

Set the model in `training` mode

```
for batch in tqdm(data_loader):
```

`data_loader` is an object that contains the data to train the model.

But what is my data?

is_offensive	text
0	"I love you"
1	"f*ck you"

But I have 180 000 rows of that

...

For training

And wednesday night at 1am

I found 160 000 new rows for testing

Few examples



"Thanks for the heads up! I will take a look!"

"Quick Question
Why do I need to join the discussion if it's resolved lol?"

"Leaders of the Christian Reconstruction movement expect a large majority of citizens worldwide eventually to accept Christ as savior. We believe in postmillennialism. Those who do not share our confidence concerning the future success of the gospel, as empowered by the Holy Spirit, believe that an earthly kingdom must be imposed by force from the top down (premillennialism), or else they do not believe in an earthly institutional kingdom at all (amillennialism).' Gary North, Political Polytheism, pp. 586-587."

That was a few non-offensive examples, you can imagine the offensive ones...

You don't need me for this part

Okay, data is good, but now what?

You need to tokenize your data

“Tokenizing a text is splitting it into words or subwords,”

"Don't you love 🤗 Transformers? We sure do."

```
["Don't", "you", "love", "🤗", "Transformers?", "We", "sure", "do."]

["Don", "'", "t", "you", "love", "🤗", "Transformers", "?", "We", "sure", "do", "."]
```

Important part

```
outputs = model(input_ids=input_ids, attention_mask=attention_mask)
```

Those params are used in transformer-based models.

- `input_ids` : Ids of each word in the input
- `attention_mask` : Which parts to pay attention to

```
optimizer.step()
scheduler.step()
```

- `optimizer.step()` : Update the weights of the model
- `scheduler.step()` : Update the learning rate

Tada 🎉

The model is training

Go grab a coffee ☕ and...

come back in a few minutes, hours, days, weeks, months, years, centuries, milleniums?



Additional usage

$0

$6.79 $30

23% of included credits used

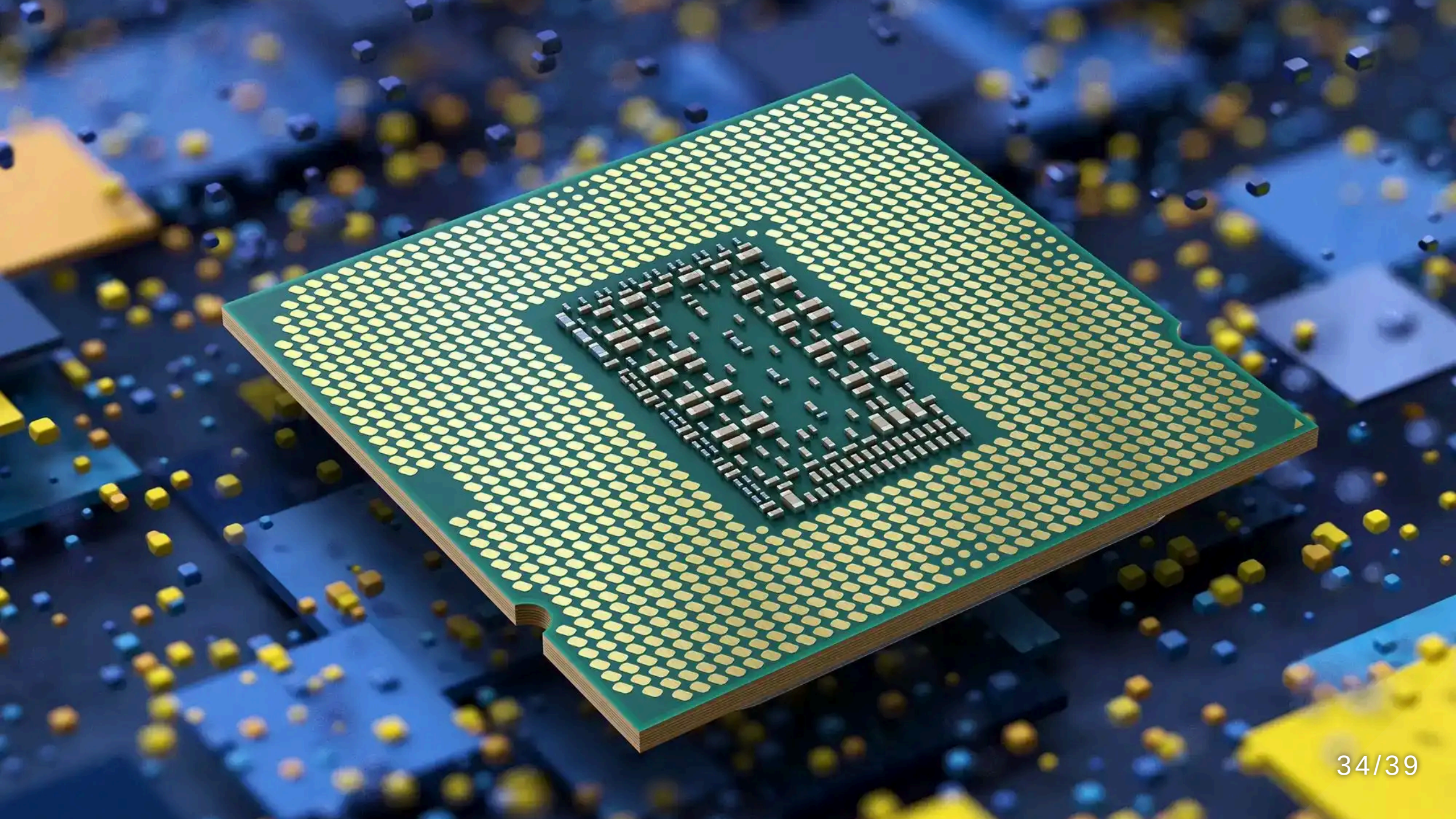
Top functions (3)

Functions are aggregated by `module_name.fn_name`.

Other usage aggregates are named in angle brackets (<>).

› train.train_on_modal	$4.49
› test.predict_on_modal	$2.29
› <image build>	$0.01

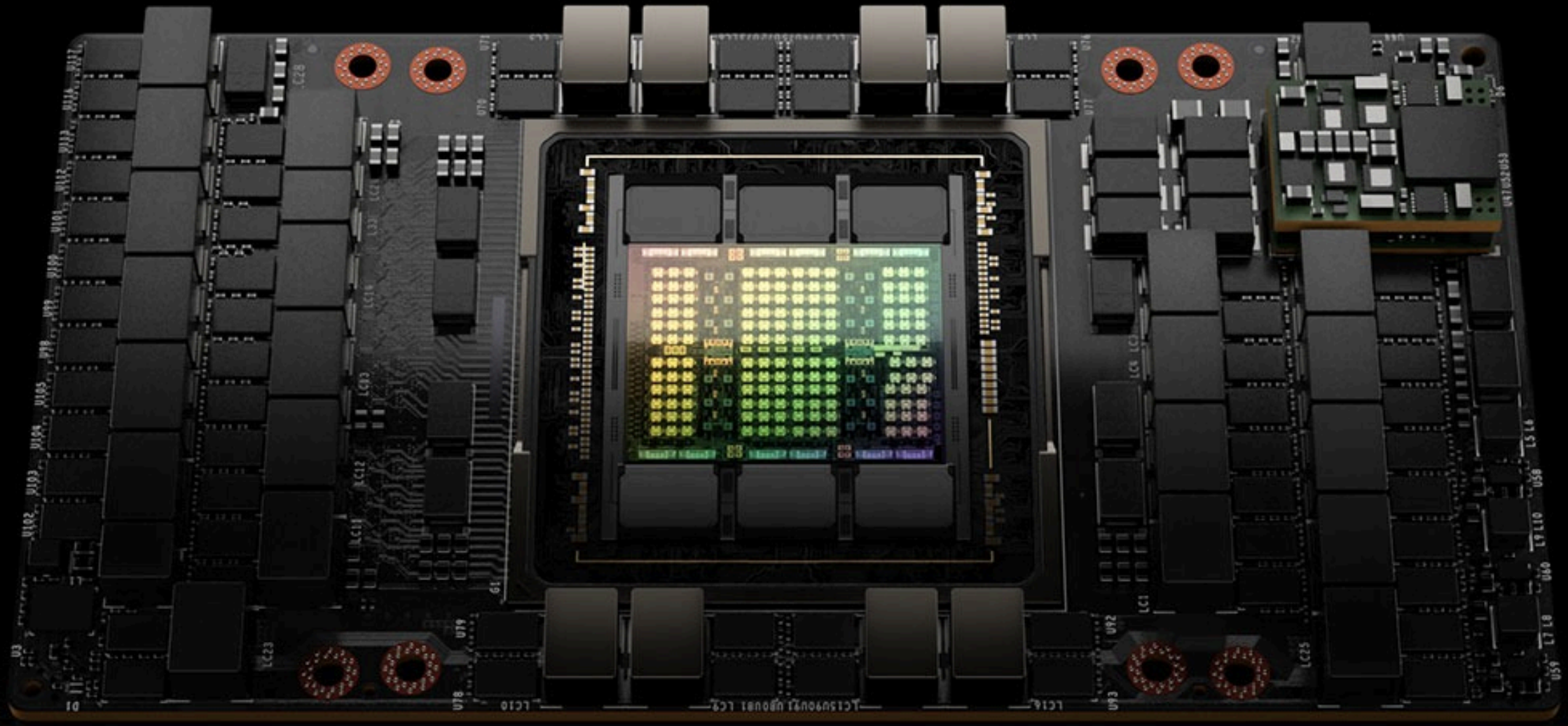
Let's run it in production



Results on a 1000 samples of the test dataset:

```
Total inference time:  0:01:02.238444 seconds
Total bad predictions:  15
Total good predictions: 985
Accuracy: 0.9847715736040609
```

~19 iterations per second



Test on the full dataset with an H100:

```
9151/159571 [01:04<14:20, 174.86it/s]
```

```
Total inference time:  0:15:12.663117 seconds
Total bad predictions:  1427
Total good predictions: 158144
Accuracy: 0.9909765783083772
```

~174 iterations per second

GPU is 9 times faster than CPU

Thanks for your attention

Once again, Spot is mad

And have fun with your code 🚀